

THE FUNDAMENTAL SYSTEMS TENSION OF PHYSICAL AI

High-Capacity Learned Generative Models ↔ Hard Real-Time Deterministic Safety Guarantees

HIGH-CAPACITY LEARNED MODELS (Cortex)

Transformers & Diffusion Policies

- **Generalization:**

Open-world semantic reasoning, novel visual grounding

- **Computational Substrate:**

Heterogeneous NPU / GPU with large DRAM footprint

- **Execution Paradigm:**

Batched tensor inference, stochastic sampling

- **Latency Profile:**

Variable 15–100 ms with long P99.9 tail spikes

- **Failure Characteristics:**

Hallucinations, out-of-distribution drift, crashes

PROPOSE
↔
PERMIT

HARD REAL-TIME DETERMINISTIC REFLEX (Enforcer)

Zero-Allocation Active-Set QP Barrier Solvers

- **Safety Guarantee:**

Strict forward invariance $h(x) \geq 0$, zero collisions

- **Computational Substrate:**

Real-Time MCU in static SRAM (0 dynamic malloc)

- **Execution Paradigm:**

Deterministic 1000 Hz loop, bounded iterations

- **Latency Profile:**

Bounded $\leq 175 \mu\text{s}$ with microsecond jitter (5 μs)

- **Failure Characteristics:**

Fail-safe hardware arrest, Category 0/1 stops